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Navigating the AI Frontier: a holistic framework for algorithmic governance and responsible innovation in organizations

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Abstract

Purpose – This study offers an integrated perspective on how organizations restructure decision-making models and adopt strategies to balance algorithmic risk, innovation, and transparency in the Artificial Intelligence (AI) era.

Design/methodology/approach – A qualitative research design was employed based on semi-structured interviews with managers, analysts, HR professionals, and public and private sector experts. Data were analysed using the Gioia Methodology to build a theoretical framework grounded in practice.

Findings – The analysis identified three key organizational dimensions: algorithmic governance (macro level), algorithmic auditing (meso level), and AI literacy (micro level). These dimensions are functionally interconnected and guided by the transversal principle of responsible innovation. Together, they form a multilevel and holistic framework for AI management in organizations. This framework also reflects how organizations dynamically sense, seize, and transform themselves in response to AI-related challenges, enriching the literature on dynamic capabilities in the digital age.

Research limitations/implications – The framework provides actionable guidance for self-assessment, policy development, and strategic investment in AI governance and training. It also highlights the relevance of new professional roles and cross-functional teams in supporting ethical and sustainable AI adoption. Since based on in-depth qualitative interviews within a European context, the findings are not intended to be statistically generalizable. Future studies could empirically test the framework in diverse sectors and national contexts, explore how organizations of different sizes operationalize AI literacy, and examine how explainability tools can be embedded into governance structures.

Originality/value – This study contributes a novel conceptualization that integrates ethical concerns, control mechanisms, and organizational culture into a unified model. Originality lies in the way individual dimensions interact, are organizationally embedded, and aligned with emerging regulatory and strategic demands. Moreover, the study expands the debate on digital transformation and hybrid decision-making structures by proposing an organizationally embedded approach to AI risk management and innovation.

Keywords Artificial intelligence, Algorithm risk, Organizational change, Organizational decision-making, Digital transformation, Innovation

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1. Introduction

The widespread adoption of Artificial Intelligence (AI)-driven algorithms for automated and semi-automated decision-making has rapidly transformed multiple sectors, extending beyond scientific research (Dos Santos *et al.*, 2023) to areas such as education, healthcare (Mulesa *et al.*, 2025), and the labour market (Morondo Taramundi, 2022). By reshaping how organizations approach decision-making, AI is fostering an increasing reliance on advanced algorithmic systems to automate complex processes and optimize strategic choices (Guercini, 2023; Cristofaro and Giardino, 2025). Algorithms, particularly next-generation machine learning (ML) models, can simplify and streamline these processes, thereby reducing individuals' cognitive and operational burdens (Halid *et al.*, 2024). However, integrating these systems is challenging (Bamel *et al.*, 2024). Just as human decisions affect individual rights and freedoms, algorithmic outcomes can exert equally profound effects. Accordingly, their pervasive use has raised concerns regarding their reliability, particularly their ability to mitigate discrimination and their impact on organizational structures and decision-making processes (Williams *et al.*, 2018). Public and private organizations alike are exposed to several risks such as opaque decision-making, limited transparency of underlying criteria, and the potential reinforcement of existing biases, all of which threaten the fairness and reliability of automated decisions (Aragona, 2021). These issues present not only technical challenges but also ethical and regulatory dilemmas, prompting scholars and policymakers to examine how best to balance algorithmic efficiency with human oversight and accountability (Tsamados *et al.*, 2021).

These concerns underscore the necessity of a structured approach to algorithmic risk management, particularly in contexts where AI algorithms are increasingly shaping organizational decision-making (Buhmann *et al.*, 2020).

Risk management serves as a cornerstone for organizations, as it protects them from disruptive impacts and ensures a safer and more sustainable innovation process in the medium and long term. At the same time, the ability to responsibly govern AI is becoming a decisive factor in maintaining long-term competitiveness, especially in sectors where digital transformation is a strategic imperative. However, while the literature has primarily focused on regulating algorithmic risk and its ex-post evaluation (Malgieri and Pasquale, 2022), less attention has been paid to understanding how public and private organizations proactively adapt their governance models to mitigate these risks while integrating AI into their decision-making processes (Koshiyama *et al.*, 2024). In particular, existing research has largely examined algorithmic risk from regulatory and compliance-oriented perspectives (Ulbricht and Yeung, 2022); in contrast, the organizational processes through which firms redesign decision-making structures, governance arrangements, and internal capabilities in response to AI adoption remain comparatively underexplored (Neiroukh *et al.*, 2025).

This perspective has shaped the emergence of the risk-based approach adopted in the AI Act (Schuett, 2024). This regulatory framework holds AI developers and users accountable for classifying applications based on their risk levels (Outeda, 2024). Nevertheless, how organizations balance algorithmic risk management with the need to innovate and maintain competitiveness remains unclear (Buhmann *et al.*, 2020). Addressing this trade-off is crucial, as organizations increasingly recognize that the strategic deployment of AI, when aligned with ethical principles, can serve as both a catalyst for innovation and a differentiator in competitive markets. Furthermore, a comprehensive framework outlining the key organizational strategies needed to manage these trade-offs is yet to be developed.

While regulatory initiatives, such as the AI Act, have significantly advanced the governance of algorithmic risk, they primarily provide normative guidelines and compliance mechanisms, offering limited insight into how organizations internally translate these principles into concrete decision-making practices and governance arrangements. Accordingly, the organizational dynamics through which firms operationalize responsible AI adoption, while simultaneously sustaining innovation, remain insufficiently theorized in the current literature.

These transformations can be interpreted through the lens of dynamic capabilities, which emphasise how organizations sense emerging technological challenges, seize opportunities for innovation, and reconfigure internal structures in response to environmental change (Teece, 2007).

Thus, the aim of this study is twofold. First, it seeks to provide an in-depth understanding of organizational transformation in response to AI. Second, it develops a conceptual framework to promote a governed and human-driven model of technological development within organizations. More specifically, the study proposes a multilevel organizational framework that explains how organizations structure and coordinate governance, auditing, and capability-building practices to manage algorithmic risk while enabling innovation. By doing so, this study highlights how responsible AI governance can serve as a strategic resource enabling organizations to innovate while safeguarding their long-term viability and market positioning.

Accordingly, this study addresses the following research questions (RQs):

RQ1. How can organizations redesign decision-making and governance models to integrate AI while mitigating algorithmic risks?

RQ2. What organizational strategies help balance algorithmic risk management with AI-driven innovation and transparency requirements?

While RQ1 focuses on how governance and decision-making structures are reconfigured to accommodate AI, RQ2 examines the broader organizational strategies through which firms balance risk management with innovation and transparency requirements. Together, these questions allow the exploration of both the structural redesign of governance arrangements and the strategic practices that support responsible AI adoption within organizations.

This study employed a qualitative approach, using semi-structured interviews with practitioners and experts involved in AI adoption across diverse organizational contexts. The collected data were analysed using the Gioia methodology (Gioia *et al.*, 2013) to inductively develop a theoretically grounded framework.

Theoretically, the study contributes to the literature by providing a conceptual foundation for future research. The analysis identifies three interrelated organizational dimensions— algorithmic governance (macro level), algorithmic auditing (meso level), and AI literacy (micro level)— that explain how organizations structure decision-making and risk management practices when integrating AI into their operations.

From a practical standpoint, it offers concrete guidance for managers and policymakers engaged in the regulation and implementation of AI in organizations by highlighting how governance mechanisms, monitoring practices, and capability development can be combined to support responsible and innovation-oriented AI adoption.

The remainder of this paper is organized as follows. Section 2 explores the impact of AI on organizational decision-making and algorithmic risk, with particular attention to the risk-based approach outlined in the AI Act. Section 3 details the research design and data analysis methodology. Section 4 presents the findings, and Section 5 discusses the key insights from the interviews. Finally, Section 6 concludes by outlining the implications for both research and practice and suggests directions for future studies.

2. Theoretical background

2.1 AI and the transformation of decision-making processes in organizations

Currently, AI plays a central role in organizational innovation, reshaping how decisions are formulated and implemented (Kaggwa *et al.*, 2024). For example, ML and natural language processing systems extend beyond automating repetitive tasks, increasingly supporting, and, in some cases, replacing human decision-making in complex and strategic contexts. Accordingly, the literature identifies two complementary uses of AI: automation and decision support. In the former, AI performs tasks that previously required human intervention, thereby reducing costs and enhancing operational efficiency (Wamba-Taguimdje *et al.*, 2020). In the latter, AI provides predictive analytics, simulations, and recommendations that enable more informed choices (Soori *et al.*, 2024). In both scenarios, AI's transformative potential lies in its ability to process vast amounts of data at unprecedented speeds and identify patterns that would be difficult for humans to detect.

This transformation has implications for organizational decision-making processes (Cristofaro *et al.*, 2025). Organizations that integrate AI into their workflows often report improvements in efficiency, reduced arbitrariness, and enhanced predictive capacity (Cockburn *et al.*, 2018).

However, this transformation is not linear; it requires organizations to seize opportunities and navigate challenges by adapting their internal structures and processes to support new decision-making dynamics, often amid uncertainty and complexity (Schoemaker *et al.*, 2018).

In particular, the literature highlights several challenges. One primary concern is decision opacity, i.e. the difficulty of understanding how AI algorithms, especially "black box" models, make their decisions (Vaassen, 2022). Moreover, issues related to data quality and representativeness are significant, as the reliability of AI systems is strongly influenced by the data on which they are trained (Bertossi and Geerts, 2020). Finally, problematic outcomes arising from automated decisions raise significant accountability concerns, as assigning responsibility becomes particularly complex amid ethical and legal implications (Dastani and Yazdanpanah, 2023).

Such challenges reveal that the introduction of AI inherently generates organizational tensions as it activates a recursive process of learning and adaptation in which existing routines and capabilities must be reconfigured through the activation of dynamic capabilities that enable organizations to sense emerging challenges, seize technological opportunities, and reconfigure internal structures (Teece, 2007; Konlechner *et al.*, 2018; Pan *et al.*, 2026).

Recent studies have explored how AI adoption reshapes organizational decision-making structures in light of these transformations (Shrestha *et al.*, 2019). Organizational theories distinguish between centralized models, in which decision-making authority is concentrated at the top, and decentralized models, in which decision-making is distributed across multiple levels and functions (Zabojnik, 2002). However, AI adoption challenges these categories, revealing that centralised and decentralised tendencies coexist. On the one hand, AI promotes centralization by enabling the aggregation of complex information into a single synthesis point (Brynjolfsson and Ng, 2023). On the other hand, it facilitates decision delegation (Candrian and Scherer, 2022) as predictive tools become accessible to mid-level or peripheral units within the organization.

These emerging configurations require the activation of dynamic capabilities at an intermediate (meso) level in order to sense organizational tensions generated by AI adoption, seize new decision-making opportunities, and reconfigure governance structures accordingly.

From this perspective, transformation is not an overarching strategy or an individual decision-making process; it emerges from the quality of interactions across organizational levels and from effective dialogue among the people involved (Salvato and Vassolo, 2018).

In this context, it is observable how, for instance, some companies have introduced hybrid roles, such as the Chief AI Officer (Schäfer *et al.*, 2022) or cross-functional data governance teams (Velinov *et al.*, 2023), which directly influence internal power dynamics and control mechanisms. Others have revised their hierarchical structures to enable faster, more iterative decision-making processes by leveraging intelligent dashboards and automated recommendation systems.

In this evolving landscape, the growing interdependence between humans and AI raises critical questions about organizations' ability to adapt to environments characterized not only by collaboration but also by inherent tensions between human and machine agency.

From this perspective, integrating AI into decision-making processes requires increasing attention not only to strategic and organizational aspects but also to the associated risks, responsibilities, and governance challenges (Krafft *et al.*, 2022).

2.2 Algorithmic risk: challenges and governance strategies in organizational decision-making

While AI offers opportunities to optimise decision-making, it can also lead to unintended negative consequences (Rodrigues, 2020). In this context, algorithmic risk – the potential error, bias, or harm arising from the use of algorithms in high-impact decision-making processes (Aragona, 2021) – emerges. It can have various forms, including automated discrimination (Žliobaitė, 2017), decision-making opacity (Chesterman, 2021), and system security and reliability issues (Hanif *et al.*, 2018), which directly affect governance practices in both private and public organizations.

This also represents a deeper managerial and structural challenge concerning organizations' ability to sense emerging failures, seize corrective mechanisms, and reconfigure their governance routines to preserve accountability and trust (Teece, 2007; Mikalef and Gupta, 2021).

Several studies have highlighted that the use of automated systems for personnel selection (Kupfer *et al.*, 2023), performance evaluation, credit management, and criminal risk forecasting can lead to unfair or even unconstitutional outcomes when the input data are biased or incomplete (Roselli *et al.*, 2019). An example is the Crime Anticipation System (CAS) used by Dutch law enforcement, which is based on historical data and statistical models. According to the data protection authority, this predictive system was deployed without adequate bias monitoring and with limited transparency (Van Schie and Oosterloo, 2020). Similarly, financial institutions utilize algorithms for fraud detection and anti-money laundering controls; however, these systems risk unfairly penalizing certain categories of users based on spurious correlations (Pombal *et al.*, 2022).

Deliveroo and Uber have demonstrated how algorithmic systems can generate discriminatory effects against workers, linking remuneration or profiling to variables influenced by gender stereotypes or behaviours due to personal circumstances (Muller, 2019; Purificato, 2021). Moreover, Amazon's AI-driven hiring system for a long time excluded female candidates (Andrews and Bucher, 2022) due to "proxy discrimination", whereby specific characteristics (such as gender) became indirectly associated with lower performance ratings in historical data.

These examples illustrate how AI adoption introduces structural risks that can undermine fundamental values, such as fairness and individual rights.

These challenges stem from the sociotechnical characteristics of AI (Latour, 2007; Graham and Rodriguez, 2021). Far from being neutral tools, algorithms reflect and often amplify the social dynamics and asymmetries embedded within the contexts in which they are designed and deployed. Consequently, algorithmic risk extends beyond a purely technical dimension, encompassing ethical, legal, and societal concerns, thereby necessitating a broader concept of organizational risk.

Recent studies emphasise that value creation and risk management are intertwined processes (Simón *et al.*, 2024). Organizations should not only respond to technical errors ex post, but also proactively reconfigure their governance systems ex ante through continuous learning and cross-functional coordination (Salvato and Vassolo, 2018).

In this broader view, algorithmic adoption introduces not only operational opportunities but also governance tensions, since it may conflict with principles such as reasonableness, proportionality, and rights protection, which are difficult to translate into computational rules (Giaccaglia, 2024).

Consequently, organizations must rethink how AI is governed and adopt, developing adaptive capacities through iterative feedback loops and productive dialogue between human and algorithmic agents (Schoemaker *et al.*, 2018).

In this vein, AI risk management becomes a multidimensional process that necessitates implementing formal governance frameworks that integrate technical tools, such as stress tests and ex-ante and ex-post controls, with regulatory principles and ethical standards. A key development in this domain is the European AI Act, a regulatory effort to classify AI applications by risk level (i.e. minimal, limited, high, and unacceptable) and establish proportionate compliance obligations (Schuett, 2024).

However, the effectiveness of such governance frameworks ultimately depends on organizations' ability to internalise them as learning mechanisms rather than mere compliance checklists (Teece, 2007; Akter *et al.*, 2023). From this perspective, the AI Act can be interpreted not only as a legal constraint but also as an external stimulus for capability development, encouraging organizations to align technical expertise, ethical reflection, and accountability for decisions within coherent governance architectures.

The literature reports an increasing adoption of mitigation strategies within organizations, ranging from creating dedicated algorithmic auditing units (Sandvig *et al.*, 2014) to implementing human-in-the-loop principles (Enarsson *et al.*, 2022) and developing internal policies for data governance. However, these practices are highly heterogeneous, with significant variations between large corporations and small and medium-sized enterprises (SMEs), as well as between the public and private sectors. For example, in public administration, algorithmic risk is linked to accountability toward citizens and the protection of fundamental rights. In contrast, in the private sector, governance tends to focus on reputational and regulatory risks, particularly those related to business performance and corporate image.

This heterogeneity confirms that algorithmic risk management is an emergent organizational capability shaped by context-specific sensing and learning processes (Helfat and Peteraf, 2015), which evolve differently depending on institutional pressures, available expertise, and organizational maturity.

Thus, algorithmic risk management should be conceived as a dynamic, path-dependent organizational capability that enables firms to learn from failure, institutionalize reflexivity, and continually reconfigure governance routines by integrating technical expertise with ethical awareness and long-term strategic learning (Teece, 2007; Pan *et al.*, 2026).

3. Methodology

3.1 Research design

The complexity of identifying and interpreting the organizational strategies that enable organizations to mitigate algorithmic risk while pursuing innovation requires a multilevel research design. Algorithmic risk exhibits several characteristics that make it difficult to observe and assess through purely quantitative indicators (Yin, 2011). First, the technical complexity of algorithms often renders their decision criteria and underlying logic opaque. This opacity limits end users' understanding and hinders the timely identification of errors or biases (Pasquale, 2015). Moreover, the variability of the contexts in which algorithms operate (for example, a CV-screening algorithm used in a technology company with high turnover versus in a public administration) makes it challenging to establish standardized criteria for risk assessment. In addition, issues of accountability further complicate the analysis, as algorithmic risk is not always attributable to a single actor but may emerge from a combination of design choices, improper use, or changes in the application context. For this purpose, a qualitative exploratory study was conducted and structured around three key phases (Figure 1). The first phase focused on selecting the data collection technique and developing an interview guide for semi-structured interviews to investigate organizational practices for implementing AI responsibly. In the second phase, qualitative data were collected through semi-structured interviews. During the third phase, the findings were analysed using the Gioia methodology (Gioia *et al.*, 2013).

This approach facilitates the identification of themes that emerge directly from the data, enabling a deeper exploration of concepts and their interconnections (Nag and Gioia, 2012).

3.2 Data collection

Data were gathered using semi-structured interviews, a particularly effective methodology for examining complex social phenomena (Sun and Zhu, 2024). All ethical standards for

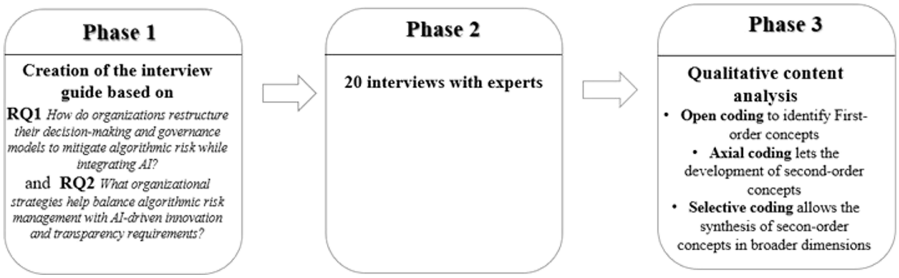

Figure 1. Research design. Source: Authors' elaboration

qualitative research were followed according to the guidelines of the researchers' institutions. Participants were informed about the objectives of the study, provided written or verbal consent before the interviews, and were assured of confidentiality and the option to withdraw at any point.

The interview guide was designed to align with the RQs and the theoretical framework. Each question (Appendix) was aligned with a specific research objective and the overall aim of the study.

For example, in response to RQ1, participants were asked questions such as: *Has the introduction of AI led to changes in internal decision-making models?* To address RQ2, participants were asked questions such as: *Does your organization implement auditing or monitoring practices for algorithms?*

The targeted questions ensured that the collected data were both relevant and comprehensive. Interviews were conducted via Zoom, Google Meet, Teams, and the telephone. Each interview lasted between 30 and 50 min, which aligns with the recommended qualitative interview practice (Robinson, 2023). The interviews were recorded, transcribed, anonymized and interpreted separately by the researchers. Respondents were asked to review the transcripts and confirm the accuracy of the information collected to ensure validity. All data were handled in compliance with the applicable data protection regulations.

Data collection and analysis followed an iterative approach in which emerging codes and themes were constantly compared with newly collected material, allowing the research team to refine the analytical focus as new insights emerged (Gioia *et al.*, 2013; Carmichael and Cunningham, 2017). Theoretical saturation was monitored across interviews by tracking the point at which additional narratives no longer resulted in new first-order concepts (Saunders *et al.*, 2018). This evaluation was conducted collaboratively among the authors through regular analytical discussions. After 16 interviews, the new ones began to reinforce the existing categories rather than introduce new conceptual elements. However, saturation was considered to be reached. To ensure robustness, two further interviews were conducted as confirmatory cases to validate the stability and coherence of the thematic structure, bringing the final sample to 20 participants (Morgan and Nica, 2020). The sample provided sufficient depth to explore the RQs and clarify the relationships among the emerging themes.

3.3 Sample and participant selection

To capture the complexity of the social phenomenon under study, various contexts in which it can be observed, such as business and academic settings, were considered in the sampling process.

Initially, a purposive sampling method was adopted (Amaturo, 2012). In the first phase, interviewees were selected from among entrepreneurs and employees whose organizations relied on AI-based services as a key resource, providing digital services, technology solutions, and consulting. These were identified through the "Osservatorio Eccellenze Italiane di

Intelligenza Artificiale" database, which is a systematic study of Italian enterprises utilizing AI to develop products and services while generating social and economic value. This database enabled access to organizations that were particularly relevant to this research.

Owing to the high non-response rate, the research design was integrated with a snowball sampling strategy (Naderifar *et al.*, 2017). Snowball sampling is widely recognised as an appropriate strategy for qualitative studies addressing emerging or complex phenomena (Parker *et al.*, 2019), particularly when the relevant population is difficult to identify in advance, and access often relies on trust-based professional networks (Sadler *et al.*, 2010).

At the end of each interview, participants were asked to suggest additional contacts who met the study criteria. This process enabled the researchers to expand the sample beyond the initial target, incorporating professionals from adjacent domains, including academic researchers, university professors, journalists, consultants, and AI ethics advisors. This snowball procedure facilitated access to individuals with specialized expertise and emerging roles that are often less visible through conventional recruitment strategies, thereby increasing the heterogeneity of professional perspectives included in the study.

Combined with purposive sampling, this approach ensured conceptual variation in line with recommendations for qualitative research (Patton, 1999).

The final sample consisted of 20 participants selected to ensure sufficient variation in terms of AI application domains, organizational roles, seniority levels, and types of institutions involved in algorithmic development and governance. Participants were drawn from both private organizations and public or academic institutions, including technology firms, consulting companies, universities, and policy-oriented organizations, reflecting variation in organizational size and degree of engagement with AI-based systems.

Descriptions of the interviewees, their different roles, and sociodemographic characteristics are provided in Table 1.

3.4 Data analysis

The collected data were examined and interpreted using qualitative content analysis, which enables the representation of complex findings in an accessible form for non-academic audiences, such as entrepreneurs and business managers (Whitney *et al.*, 2010).

In accordance with the Gioia methodology (Gioia *et al.*, 2013), the data analysis followed an iterative, inductive progression, enabling the emergence of theoretical insights directly from the data. The process, as detailed below, was structured in three stages: identification of first-order concepts based on participants' own terms and meanings; abstraction of second-order themes, reflecting interpretative patterns across interviews; and consolidation of aggregate dimensions, representing higher-level conceptual structures.

Open coding: The transcribed interviews, considered units of analysis, were examined to identify first-order concepts using informant-centric terms and codes. At this stage, the analysis focused on capturing participants' own expressions and meanings in order to preserve the empirical richness of the data.

Axial coding: The analysis of relationships among first-order concepts led to the development of second-order concepts that were more researcher-driven and theoretically informed. These second-order themes were developed through a continuous process that combined deductive reasoning with inductive exploration. Through this process, conceptually related codes were progressively grouped into broader analytical categories.

Selective coding: Second-order themes were thus synthesised and organized into broader dimensions. At this stage, the analysis moved toward theoretical abstraction by identifying the underlying conceptual logic linking related themes.

For instance, during open coding, informants frequently described how AI tools encouraged employees to broaden their understanding of work processes and their interdependencies:

Table 1. Description of the sample

	Job position	Sector	Organization type	Sex	Age	Code
1	Entrepreneur	Private	Tech SME	Male	56	E1
2	Entrepreneur	Private	Tech SME	Male	52	E2
3	Data scientist	Private	IT consulting firm	Female	35	DS1
4	Digital Consultant	Public	Public body	Female	37	DC1
5	Entrepreneur	Private	Tech Start-up	Male	29	E3
6	Data science Team leader	Private	IT consulting firm	Male	37	M1
7	Entrepreneur	Private	Tech firm	Male	47	E4
8	Researcher	Public	University	Female	32	R1
9	Manager (Director of AI)	Private	Software company	Male	45	M2
10	Data scientist (R programmer)	Private	Software company SME	Male	27	DS2
11	Entrepreneur	Private	Digital Start-up	Male	34	E5
12	Data scientist	Private	Digital technology SME	Male	28	DS3
13	Principal Data scientist	Private	Software company	Female	39	DS4
14	AI and Data strategy advisor	Private	IT consulting firm	Male	39	DC2
15	Chief data scientist	Private	Digital technology SME	Male	43	C1
16	Chief AI officer	Private	Software company	Male	40	C2
17	Senior lecturer and research affiliate	Public	University	Male	46	R2
18	Business and technology journalist	Private	Media organization	Female	38	J1
19	AI engineering coach	Private	IT consulting firm	Male	37	C3
20	Researcher in big data analytics	Public	University	Male	36	R2

Source(s): Authors' elaboration

> AI pushes us to adopt an end-to-end view of workflows, helping people understand how their actions affect the organization as a whole.

> Creating a more collaborative environment leads employees to extend the use of AI solutions to other internal and external processes, opening new growth opportunities.

These statements were initially coded as first-order concepts such as "greater process integration", "expanded collaboration", and "extended use of AI across tasks". During axial coding, these codes were grouped into a second-order theme reflecting the reconfiguration of work practices toward more interconnected and collaborative forms of decision-making. Finally, in the selective coding phase, this theme contributed to an aggregate conceptual category, capturing broader organizational adaptations associated with AI-enabled transformation.

To enhance trustworthiness and rigor, multiple forms of triangulation and validation were adopted (Riazi *et al.*, 2023):

(1) Investigator triangulation: Each researcher independently coded the transcripts, and discrepancies were discussed in iterative comparison sessions to refine category boundaries, reduce individual bias, and enhance the reliability of the analysis (Patton, 1999; Hanson-DeFusco, 2023).

(2) Participant validation (member checking): Preliminary interpretations and emergent categories were shared with a subset of participants through a member validation workshop to assess whether the analytical representations aligned with their experiences (Lincoln *et al.*, 2011; Alessi and Kahn, 2023).

(3) Consideration of alternative interpretations: Throughout the coding process, the research team systematically examined disconfirming evidence and rival explanatory patterns, documenting analytic decisions through an audit trail (Carcary, 2009). Coding disagreements and alternative conceptual groupings were discussed iteratively before defining second-order themes (Varpio *et al.*, 2017). This reflexive

comparison helped avoid premature closure and enhanced the credibility and internal validation of the emergent framework (Raven, 2007).

Following this process, the findings were structured within a framework designed to accurately represent the dimensions and subdimensions of the investigated constructs. This framework helps to identify the activities, practices, and behavioural characteristics of a critical approach to AI-driven decision-making that mitigates potential negative impacts on organizations and individuals.

4. Results

The analysis of in-depth interviews enabled the construction of a data structure (Figure 2) organized into three aggregate dimensions. These dimensions correspond to the conceptual categories inductively derived from the data: algorithmic governance, algorithmic auditing, and AI literacy (Table 2). The following sections analyse these three dimensions through a narrative supported by significant quotes from the interviews.

4.1 Algorithmic governance

The first aggregate dimension concerns the governance and strategic management of AI systems within organizations. The increasing pervasiveness of algorithmic tools requires companies to rethink how decisions are made, who is responsible, and how risks are anticipated and managed. Rather than treating AI as a purely technical matter, interviewees stressed the importance of integrating governance mechanisms that cut across organizational layers and functions. These insights indicate that AI adoption prompts organizations to reconsider the distribution of decision authority and institutional arrangements through which algorithmic risks are anticipated and governed.

4.1.1 Strategic involvement and leadership awareness. A key theme emerging from the interviews was the strategic role that top management must play in the adoption of AI. Far from being a purely operational domain, AI governance requires high-level involvement, particularly in contexts where algorithmic decision-making may have significant ethical, legal, or reputational implications.

As E1 stated, “top management cannot simply delegate AI-related matters to technical teams. They must at least understand the primary risks and actively participate in defining internal policies”. E2 echoed this, noting that “AI is not just a tool; it is a strategic shift. If leaders do not engage directly, we risk building systems that are misaligned with our core values”. These statements suggest that AI governance increasingly requires strategic oversight at the executive level, where leadership involvement becomes essential to ensure that algorithmic systems remain aligned with organizational objectives and ethical commitments.

Several organizations have addressed this challenge by establishing new governance structures. “We have created an internal ethics committee composed of management representatives, legal experts, and AI specialists, to oversee the development of the most critical models”, explained C2. This example illustrates how organizations institutionalize algorithmic governance by creating multi-stakeholder oversight structures that embed ethical considerations in the development of AI systems and align technological decisions with broader organizational and societal expectations.

4.1.2 Internal policies and procedural safeguards. Beyond strategic oversight, AI governance is operationalised through the implementation of internal rules and procedural safeguards. These include transparency requirements, documentation practices, and dedicated protocols for managing the risks associated with automated systems.

As DS1 reported, “We have incorporated into our corporate policy the obligation to disclose the use of AI systems in all automated decision-making processes, especially those involving external users or impacting employees”. DS2 emphasises the role of transparency as a precondition for trust. “Without clear guidelines and documentation, people simply won’t

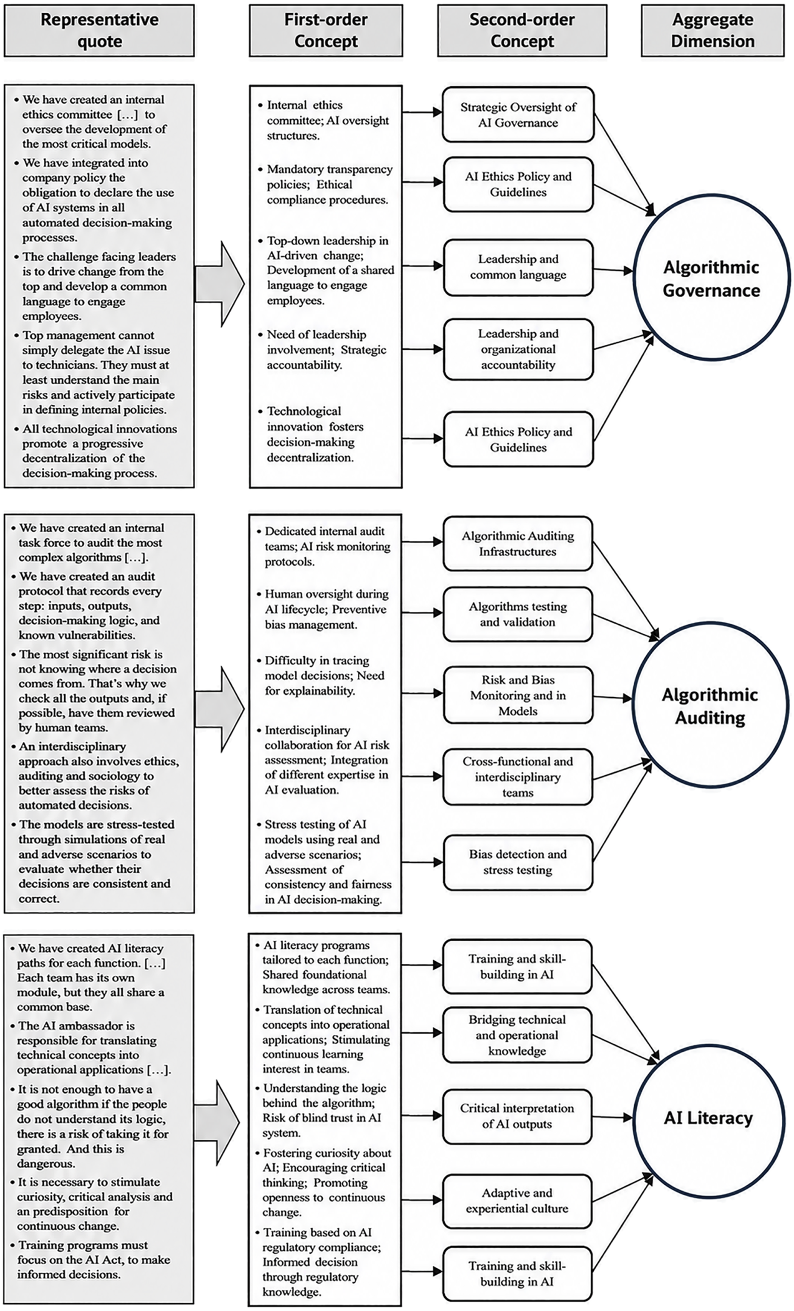

Figure 2. Data structure (Gioia methodology). Source: Authors' elaboration

trust what the system recommends, whether it's a loan decision or a performance evaluation". These observations suggest that transparency and documentation practices serve as

Table 2. Summary of dimensions, themes and representative quotes

Representative quote	First-order concept	Second-order theme	Aggregate dimension
We have created an internal ethics committee [. . .] to oversee the development of the most critical models	Internal ethics committee; AI oversight structures	Strategic Oversight of AI Governance	Algorithmic Governance
Top management cannot simply delegate AI-related matters to technical teams. They must at least understand the primary risks and actively shape internal policies	Leadership awareness of AI risk; Strategic involvement	Leadership Engagement in AI Strategy	
We integrated into company policy the obligation to declare the use of AI systems in any automated decision that may affect employees or external stakeholders	Mandatory transparency policies; Accountability rules	AI Ethics Policy and Governance Guidelines	
AI fosters a shift toward an end-to-end view of processes, integrating previously separate activities and clarifying the organizational impact of decisions	End-to-end organizational restructuring; Workflow integration	Organizational Alignment for AI Adoption	
Creating a collaborative environment encourages employees to extend the use of AI to other processes, generating new opportunities	Team cooperation; Diffusion of AI practices across units	Cross-Functional AI Coordination	
We created an internal task force to audit the most complex algorithms [. . .] with stress-test simulations and periodic evaluations	Dedicated internal AI audit teams; Stress-testing routines	Algorithmic Auditing Infrastructures	Algorithmic Auditing
The greatest risk is not knowing the origin of an automated decision. That's why every critical output is reviewed by human evaluators	Human-in-the-loop oversight; Prevention of opaque decision-making	Human Oversight and Bias Monitoring	
We developed an audit protocol that documents inputs, outputs, decision logic and vulnerabilities to ensure traceability over time	Audit documentation systems; Traceability practices	Algorithm Testing and Validation Protocols	
During AI model reviews, we conduct adverse scenario simulations to identify potential biases and vulnerabilities	Scenario-based robustness testing; Bias stress tests	Risk Anticipation and Mitigation Practices	
A good algorithm is not enough. If users see it as infallible, they will stop questioning the decision process	Critical awareness of model limitations; Avoidance of automation bias	Ethical and Reflexive AI Use	AI Literacy
We developed AI literacy programs for every function, all based on a shared foundation but customized by role	Organization-wide training; Role-based modules	Training and Skill-Building in AI	
The AI Ambassador helps translate technical concepts into everyday work practices and supports colleagues in applying them responsibly	Internal facilitation of knowledge sharing; Peer-driven learning processes	Bridging Technical and Operational Knowledge	
Moving from digital illiteracy to AI awareness requires a cultural shift, not merely a technical one	Cultural transformation; Shared understanding of AI logic	Organizational Culture Shift for AI Integration	
Fragmented training programs create silos: AI literacy must be consistent to enable dialogue and coordination across functions	Need for unified training frameworks; Avoiding siloed learning	Cross-Functional Learning Alignment	

Source(s): Authors' elaboration

organizational safeguards, making algorithmic decision-making processes more accountable and interpretable for both internal and external stakeholders.

This proactive stance is increasingly viewed as necessary to anticipate both regulatory developments and reputational risks. For example, participants referenced the importance of aligning with forthcoming EU rules, such as the AI Act, and internalising the principles of responsible AI. “We’ve decided to integrate the principles of upcoming regulation – like explainability and fairness – into our development process, even if not yet mandatory” (DC1). This illustrates how organizations proactively translate emerging regulatory principles into internal governance routines, aligning technological experimentation with evolving compliance expectations.

4.1.3 Towards a culture of shared responsibility. Finally, several respondents suggested that governance is not only a matter of formal policies but also of organizational culture. In their view, fostering shared responsibility across departments is essential to ensure that AI systems are deployed in a socially responsible and legally compliant manner. “Algorithmic systems are not just technical tools, they’re powerful and potentially harmful creators of new complexities”, argued E3. “This requires firm accountability from all users, at every stage of the lifecycle” (R1). M1 emphasised the need to avoid silos. “We need cross-functional teams who understand both the ethical and operational sides of AI. Otherwise, we risk losing control of how these systems evolve”. These insights suggest that effective AI governance increasingly depends on the diffusion of responsibility across organizational roles, supported by cross-functional collaboration and lifecycle accountability rather than being confined to technical specialists alone.

4.2 Algorithmic auditing

The second aggregate dimension concerns the formalisation of oversight mechanisms and audit practices to monitor, test, and validate the functioning of AI systems. The participants acknowledged that algorithmic tools, especially when applied to complex and high-stakes decisions, inevitably introduce risks related to data quality, model bias, and output opacity. In this context, auditing emerges not as a bureaucratic requirement but as a central pillar of responsible AI use. In contrast to governance structures that define strategic oversight, auditing practices focus on the continuous monitoring, testing, and validation of AI systems in everyday organizational operations.

4.2.1 Risk awareness and internal audit mechanisms. Several respondents described the establishment of internal structures dedicated to reviewing and validating AI systems. This typically involves cross-functional teams tasked with conducting simulations, stress tests, and performance monitoring.

“We have established an internal task force for auditing the most complex algorithms: we conduct periodic tests and simulations with adverse scenarios to assess their robustness”, shared M2. This example illustrates how organizations are formalizing internal auditing routines that systematically test algorithmic robustness before and during deployment.

These practices are particularly valuable in environments in which systems are used to support or automate decisions with ethical or legal implications. DS3 emphasised, “You can’t just assume the model works, you need to pressure-test it under different conditions and datasets”.

This observation highlights a shift from passive reliance on algorithmic outputs toward continuous testing practices designed to detect potential failures or unintended consequences.

Moreover, these audit mechanisms are often viewed as reactive tools for error detection and proactive instruments for improving model performance over time. In this sense, auditing practices function both as risk-mitigation mechanisms and as organizational learning tools that support the iterative refinement of AI systems.

4.2.2 Addressing opacity and bias. Another key concern emerging from the data is the difficulty in interpreting and explaining the outputs of increasingly complex AI models, particularly those based on deep learning architectures. In response, the participants described various strategies for identifying and mitigating bias, ranging from human-in-the-loop reviews to post-hoc explainability tools.

"The greatest risk is not knowing the origin of a decision. For this reason, we review all outputs and, when possible, have them examined by human teams" (DS4). This illustrates how organizations combine algorithmic outputs with human oversight to maintain interpretability and accountability in automated decision-making.

Others pointed to the role of domain experts in interpreting model behaviour and identifying problematic patterns. "If a model keeps rejecting the same profile, someone needs to ask why. Otherwise, we risk institutionalising discrimination without realising it" (DC2). These insights indicate that bias detection relies not only on technical validation but also on contextual interpretation by domain experts who can identify problematic decision patterns.

In this sense, auditing practices extend beyond purely technical checks, enabling organizations to critically assess the social and organizational implications of algorithmic decisions.

4.2.3 Documentation and replicability. A final theme within this dimension relates to the need for the traceability and replicability of auditing procedures. Participants emphasised that consistent documentation of models' input-output logic, vulnerabilities, and revision history was essential to maintaining long-term oversight, especially in organizations where models evolve rapidly or are modified by different teams.

"We have developed an auditing protocol that documents every step: inputs, outputs, decision-making logic, and known vulnerabilities. This is the only way to keep track and improve over time," (C1). This example illustrates how documentation practices institutionalize algorithmic accountability by enabling the systematic tracing and evaluation of model behaviour over time.

C3 added, "Without documentation, it's impossible to understand what changed from one version to the next, especially if the person who built the model leaves the company". This observation highlights how documentation practices preserve organizational memory and ensure the traceability of algorithmic systems over time, reducing risks associated with personnel turnover or model evolution and supporting accountability throughout the lifecycle of AI models.

4.3 AI literacy

The third aggregate dimension relates to the cultural and cognitive infrastructure that enables individuals across an organization to meaningfully engage with AI. While algorithmic systems can technically function in isolation, their responsible integration into organizational processes depends on the human capacity to interpret, contextualize, and question their outputs.

Across interviews, respondents highlighted a significant gap in AI literacy, not only in terms of technical proficiency, but more fundamentally, in the ability to critically assess what algorithms do, how they work, and why they matter. This gap was described as a major barrier to the responsible adoption of AI within everyday organizational practices.

4.3.1 Awareness of cognitive risks. Several participants warned that the absence of a shared understanding of AI systems could lead to overreliance or blind trust in automated outputs. "A good algorithm is not enough: if users don't understand how it works, they may see it as infallible, which is dangerous", noted J1. This kind of cognitive risk, which R2 called "automation complacency", was described as particularly problematic in operational contexts where employees defer to machine recommendations without questioning them. These insights indicate that limited AI literacy can foster cognitive dependence on algorithmic outputs, leading employees to rely on automated recommendations without adequately questioning their underlying logic.

This highlights the growing recognition that AI risk is not only technical or organizational but also epistemic, rooted in how people make sense of machine logic.

4.3.2 Training programs and cross-functional learning. To address these issues, many organizations have introduced structured training programs aimed at building a baseline for

AI-related knowledge across departments. These initiatives often aim to demystify AI concepts and foster interdepartmental collaboration.

"We have developed AI literacy programs for every function, starting with fundamental concepts to concrete use cases. Each team has its own module, but they all share a common foundation," explained E4, who was involved in program design.

C1 added, "The goal is not to turn everyone into data scientists, but to make sure they can ask the right questions and understand the implications". So, organizations are increasingly framing AI literacy as a collective, interpretive capability. Rather than focusing solely on technical training, these initiatives aim to equip employees to critically interpret algorithmic outputs and engage with AI systems across different organizational functions.

4.3.3 Internal champions and AI ambassadors. A particularly innovative practice reported by some organizations involves the introduction of AI Ambassador roles, individuals who act as knowledge translators, facilitators, and promoters of ethical AI use within teams. "The AI Ambassador translates technical concepts into operational implications for the team and helps maintain ongoing engagement with training initiatives" (E5). These roles often emerge informally but are increasingly being formalised as part of broader transformation strategies. According to R1, "It's about building a bridge between tech teams and the rest of the organization, someone who speaks both languages and can mediate". These insights show how organizations are introducing intermediary roles that translate technical knowledge into operational practices, enabling communication between technical and non-technical actors. In this way, AI literacy becomes embedded within everyday organizational routines rather than remaining confined to specialised technical domains.

5. Discussion

The analysis of semi-structured interviews provides empirical insights that directly address the research questions of this study, showing how organizations redesign governance and decision-making structures to integrate AI (RQ1) while simultaneously developing strategies that balance algorithmic risk management with innovation and transparency (RQ2). In particular, the analysis reveals a shift toward more distributed decision-making models guided by conscious leadership and flexible governance structures.

In fact, AI has emerged as an accelerator of delayering processes (Rudko *et al.*, 2021), fostering the involvement of cross-functional figures (Ahmad *et al.*, 2023), and leading to the creation of interdisciplinary task forces (Hine and Barnaghi, 2024). Moreover, the analysis shed light on formal oversight mechanisms, such as ethics committees and transparency policies, as well as new bridging roles, such as AI ambassadors. Also, ethics guidelines, internal supervision protocols, and awareness of explainability practices serve as formalization mechanisms that enable organizations to maintain control while simultaneously fostering innovation (RQ1).

The results also highlight three interrelated strategies that help balance transparency, ethics, and experimentation (RQ2). The first is continuous auditing (Aragona, 2021), which allows organizations to assess the impact of AI models, even during the experimental phases. The second is widespread AI education, which prevents uncritical or technocratic approaches (Furey and Martin, 2019). The third is the adoption of a systemic perspective, where risk is not confined to regulatory compliance but is embedded within organizational culture as a tool for steering innovation.

Within this emerging organizational configuration, three interrelated dimensions become particularly salient: algorithmic governance, algorithmic auditing, and AI literacy.

These concepts capture distinct but complementary organizational functions: governance provides the strategic and normative architecture for AI adoption, auditing translates this architecture into operational practices of monitoring and verification, and AI literacy supplies the cognitive and cultural conditions necessary for their effective and responsible enactment.

In this way, organizations can advance the ethical adoption of AI while strengthening organizational agility and resilience within increasingly data-driven ecosystems.

Algorithmic governance (macro level) emerges as a strategic foundation (Coglianese and Lehr, 2019): leadership engagement, policy development, oversight committees, and the construction of a shared language ensure that AI is not treated as a technical matter but as a central theme in organizational strategy.

Governance serves as a framework within which norms, roles, standards, and inter-institutional relationships evolve, delineating the ethical and regulatory perimeter of AI systems (Fortes *et al.*, 2022). This draws on the organization's sensing capability to detect signals of technological and regulatory change, interpret ethical tensions, and anticipate external expectations (Thiel *et al.*, 2012; Lescrauwaet *et al.*, 2022).

Such anticipatory capacity also relies on the broader institutional environment, in which, for instance, the active involvement of public agencies is valuable, as they can fulfil roles related to oversight, training, inspection, and regulatory support for organizations (Valle-Cruz *et al.*, 2019). Thus, this implies that algorithmic governance should encompass both internal mechanisms (i.e. policies, ethical codes, and committees) and external references (i.e. regulations, standards, and regulatory authorities) (Engstrom and Ho, 2021). By doing so, organizations could be better equipped to design governance architectures that are not only compliant but also adaptive and forward-looking.

This dimension underscores the trend toward flexible formalization of governance, where decision-making structures evolve in response to technological risk (Liu *et al.*, 2021), adopting distributed models in which algorithmic governance becomes the cornerstone of a new equilibrium between authority and shared responsibility (Stone *et al.*, 2020) (RQ1).

Algorithmic auditing (meso level) reflects organizational awareness of the inherent limitations of AI models (Hassija *et al.*, 2024). It encapsulates an organization's seizing capability to reconcile performance with precaution (Lengnick-Hall and Beck, 2016) (RQ2) through the timely design and implementation of internal structures such as audit protocols, impact assessments, and explainability tools.

In this way, organizations translate risk signals into tangible practices of control and trust-building, thereby institutionalising ethical reflection within the innovation process while preserving organizational agility and performance.

Thus, risk management does not become merely a defensive barrier but also a strategic lever for building more robust and enduring innovation (Lv *et al.*, 2018), enabling organizations to pursue experimentation sustainably while ensuring accountability and transparency toward both internal and external stakeholders (Akinrinola *et al.*, 2024).

Finally, AI literacy (micro level) is an enabling pillar, embodying the cognitive and operational tools needed to engage critically with these technologies (Almatrafi *et al.*, 2024).

It is grounded in an organization's reconfiguring capability, which enables firms to adapt and realign internal competencies, routines, and structures in response to AI-driven transformation (RQ1, RQ2), fostering broader awareness of AI's implications for decision-making processes and thereby promoting its ethical and strategic use.

The results demonstrate that a shared understanding of AI – among both technical and managerial teams – is essential for mitigating risks, maximizing AI's potential, and supporting its responsible adoption (Schiff *et al.*, 2020). At this level, skill acquisition concerns the understanding of the underlying logic of AI, the data that fuels it, and the consequences of the recommendations it generates (Von Eschenbach, 2021). Thus, AI literacy becomes a strategic asset that enhances human capital and strengthens organizational resilience (Li and Kim, 2024).

Organizations that invest in widespread literacy demonstrate a greater ability to restructure their decision-making models and drive innovation (Giustini and Dastyar, 2024) with a strong commitment to transparency, ultimately enhancing their overall digital maturity.

These three dimensions create an integrated framework (Figure 3) that balances experimentation and control, innovation and transparency, and efficiency and responsibility.

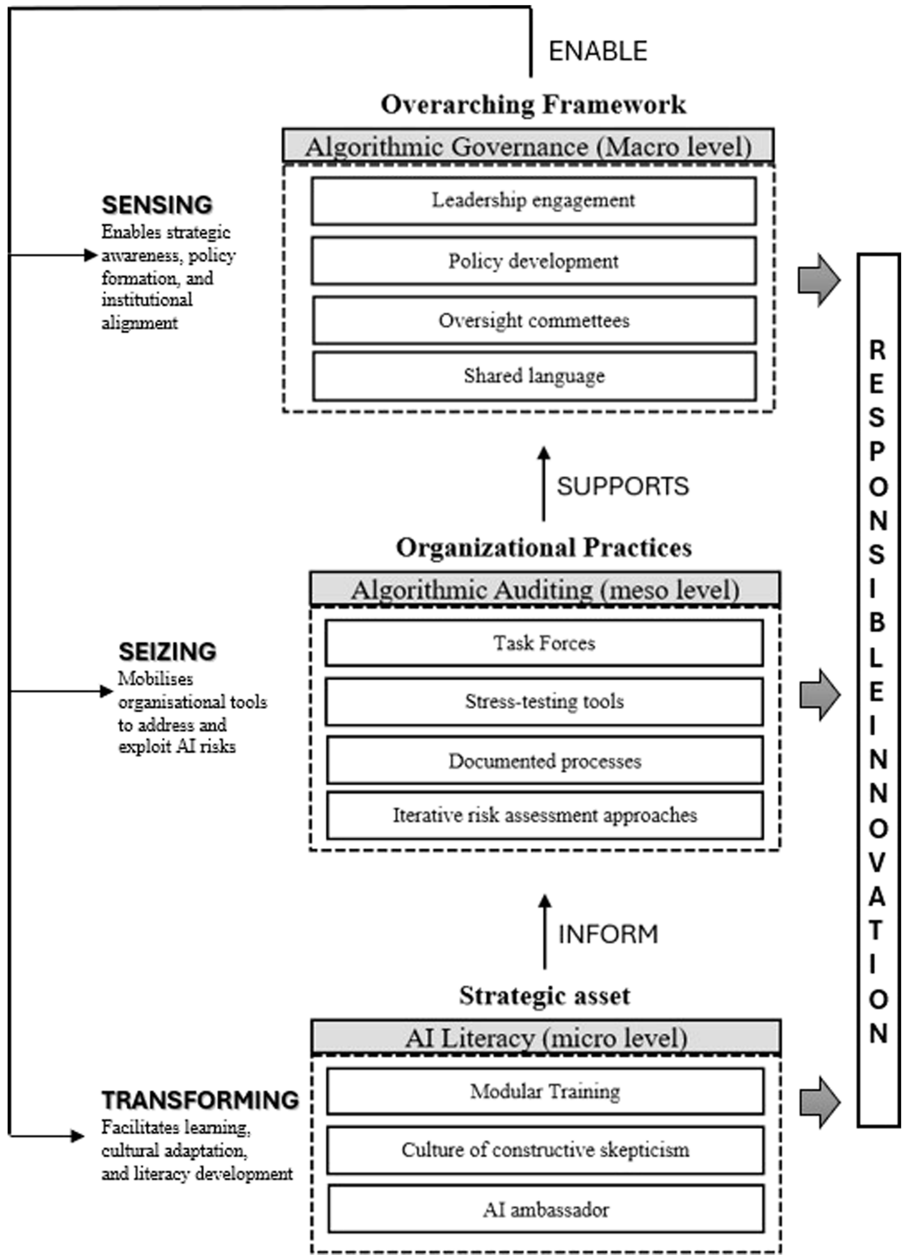

Figure 3. A multilevel organizational capability framework for responsible AI governance. Source: Authors' elaboration

The framework developed in this study focuses on the organizational mechanisms through which these principles are enacted in practice. Rather than proposing an additional regulatory or technical model—such as the EU AI Act or the NIST AI Risk Management Framework

(2024)—the framework advanced in this study adopts a multilevel logic of adaptation. This perspective reflects the need for organizations to translate regulatory principles into concrete decision-making routines and innovation processes, while dynamically realigning structures, processes, and competencies in response to AI-driven transformations. Consistent with Teece's (2007) theory of dynamic capabilities, this adaptive logic emphasises the organization's ability to sense external changes, seize emerging opportunities, and transform internal routines, thereby governing technological transitions while safeguarding ethical integrity, equity, and accountability.

To achieve this balance, organizations must strategically integrate governance and auditing mechanisms into their innovation processes. Rather than constituting a constraint, algorithmic governance functions as a structural enabler of sustainable AI-driven innovation. By establishing clear oversight structures and ethical committees, organizations can foster a culture of accountability in which technological advancements remain aligned with evolving organizational values and external expectations. In this context, the ability to proactively identify emerging risks and societal demands becomes essential for informing adaptive policy design and institutional readiness (Shafizadeh, 2024).

A cross-cutting principle that implicitly emerges from the results is responsible innovation, a vision of technological progress attentive to ethical, social, and organizational implications (Von Schomberg, 2013). Operationalizing this vision requires continuous refinement of internal capabilities, particularly those that support agile adaptation to change (Zollo *et al.*, 2016; Mueller-Saegebrecht and Walter, 2025). This includes not only formal auditing infrastructures but also shared routines for evaluating algorithmic performance under uncertainty and ambiguity.

In this vein, AI literacy transcends technical training and becomes a strategic asset, enabling organizations to reconfigure decision-making routines and enhance their absorptive capacity (Malik, 2023). Embedding literacy programs within long-term strategies allows all stakeholders, including non-technical personnel, to critically engage with AI, recognize its limitations, and contribute to its responsible deployment (Eitel-Porter, 2021). By institutionalizing ethical principles and interpretative awareness across the AI lifecycle – from model development to organizational uptake – organizations are better positioned to evolve in data-rich environments while preserving accountability and transparency (Seppälä *et al.*, 2021).

6. Conclusion

This study examined how organizations integrate AI into decision-making while managing algorithmic risks using a qualitative approach (Gioia *et al.*, 2013) to analyse interviews with managers and experts from both the public and private sectors.

The analysis identified three interrelated organizational strategies – algorithmic governance, algorithmic auditing, and AI literacy – that operate not as isolated levers, but as mutually reinforcing mechanisms embedded across the macro, meso, and micro levels. The overarching principle connecting these strategies is responsible innovation, understood as the ability to balance experimentation, ethics, and control (Von Schomberg, 2013). The findings underscore the need for organizations to adopt a multi-level approach to algorithmic transformation. This embedded and interrelated perspective is conceptualised in an inductively derived framework that, unlike existing models such as the EU AI Act or the NIST AI Risk Management Framework (2024), moves beyond normative or procedural guidance to show how organizations translate abstract principles into situated operational practices.

By focusing on the organizational embedding of AI risk management, the framework illustrates how ethical, strategic, and learning mechanisms converge to build internal infrastructures that sustain responsible and resilient innovation (Von Schomberg, 2013). Accordingly, it offers both theoretical and practical contributions to understanding and supporting the transformation of organizations in the age of AI.

6.1 Theoretical implications

From a theoretical perspective, this study proposes an integrated model that captures the interdependence of governance, risk, and digital skills, dimensions often addressed separately in previous research (Mäntymäki *et al.*, 2022; Cheatham *et al.*, 2019; Morandini *et al.*, 2023).

Additionally, the developed framework enriches the debate on dynamic capabilities (Cristofaro and Lovallo, 2022), suggesting that AI management requires cognitive, regulatory, and cultural infrastructure (Wamba *et al.*, 2024).

Furthermore, by tracing how organizations sense, seize, and reconfigure their response to algorithmic risk, the framework offers a dynamic perspective on how AI-related capabilities evolve across strategic, procedural, and cognitive domains (Gama and Magistretti, 2025).

This work also contributes to studies on automated and hybrid decision-making (Jarrahi, 2018) by offering a perspective centred on organizational adaptation, expanding the discussion of algorithmic governance models (Rudko *et al*., 2021), and proposing a novel synthesis between the need for control and the drive for innovation.

6.2 Practical implications

From a practical standpoint, the proposed framework can guide strategic actions for executives, managers, HR leaders, innovation officers, and policymakers. By integrating three complementary organizational dimensions, algorithmic governance, algorithmic auditing, and AI literacy, it provides a structured roadmap for the responsible adoption of AI. In this framework, governance establishes strategic oversight and accountability structures; auditing enables monitoring and evaluation of AI systems; and AI literacy develops the capabilities needed to interpret and critically engage with algorithmic outputs. Moreover, the framework can serve as a self-assessment tool for evaluating organizations' AI management maturity, provide a foundation for developing targeted organizational policies on governance, auditing, and training, and act as a starting point for defining strategic investments, both technological and cultural. It can also be adapted to different organizational contexts. For instance, SMEs may prioritise essential auditing checklists and introductory training modules to build foundational literacy. In contrast, multinational or public organizations may focus on formal governance structures, such as ethics committees, Chief AI Risk Officers, or explainability dashboards, to make AI decisions more interpretable to non-technical stakeholders.

Furthermore, the model facilitates the formation of cross-functional teams capable of managing AI ethically and responsibly (Ahmad *et al*., 2023). It also highlights the opportunity to introduce new professional roles, such as AI Literacy Ambassador or Chief AI Risk Officer (Schäfer *et al.*, 2022), while promoting structured collaboration among often disconnected organizational areas, including IT, HR, compliance, and communication.

The findings also suggest that managers should not only trust algorithmic outputs but also develop understanding of how these outputs are generated. This calls for greater attention to AI explainability as a managerial capability, especially for those responsible for translating AI systems into strategic value. In this regard, raising awareness about techniques such as SHapley Additive exPlanations (SHAP) or Local Interpretable Model-Agnostic Explanations (LIME) can be particularly valuable. While these tools do not require advanced technical expertise, they provide concrete support for managers seeking to critically interpret AI-driven decisions, engage meaningfully with data scientists and auditors, and foster accountable use of algorithmic systems across the organization.

In the broader European context, the proposed framework can also support organizations in operationalizing key regulatory expectations around transparency, accountability, and ethical oversight. By bridging technical and managerial domains, it contributes to building an internal infrastructure that aligns with external frameworks, thereby enabling compliance and innovation while strengthening organizations' ability to compete in rapidly evolving markets, where trust, adaptability, and responsible innovation are key differentiators.

6.3 Limitation and future research directions

Despite the value of the findings, this study has some limitations. Qualitative and interpretative analyses, which are based on a purposive sample of experts, do not allow for generalization. In addition, because the data rely on expert interviews, responses may be subject to respondent bias, including social desirability or elite bias, whereby participants may emphasise best practices or normative expectations when discussing AI governance and risk management (Bergen and Labonté, 2020). However, several methodological safeguards, such as investigator triangulation, iterative coding comparisons, and participant validation, were employed to mitigate these risks and enhance the credibility of the analysis. Moreover, focusing on the European context may limit its applicability in different cultural or regulatory environments. Finally, this study examined the perspectives of actors without a direct observation of decision-making processes or technological implementation. These limitations open avenues for further research to test and refine the proposed framework across diverse empirical settings. Comparative studies across regulatory contexts may shed light on how national frameworks shape the translation of ethical and governance principles into practice. Future work could also examine how organizations of different sizes and sectors operationalize AI literacy in relation to their innovation capacity and risk exposure. Additionally, exploring whether explainability tools are integrated into internal governance and auditing routines would help clarify how such practices strengthen accountability and support informed decision-making. Addressing these questions would further consolidate the framework and contribute to a cumulative research agenda on the organizational capabilities required to responsibly manage algorithmic transformation.

Appendix
Interview sketch

Section 1 – Introduction and Context

(1) Can you describe your role and your current level of involvement in digital transformation processes and AI adoption within your organization?

(2) How has your organization started integrating artificial intelligence solutions? Are there any priority areas?

Section 2 – Decision-Making Structures and Governance

(3) Has the introduction of AI led to changes in internal decision-making models? If so, what are they?

(4) How are decisions regarding the implementation or use of algorithms made? Who is involved?

(5) Are there specific roles, committees, or procedures in place to oversee the use of AI?

(6) Can you provide examples of cases where a decision made by an automated system needed to be revised or overridden?

Section 3 – Algorithmic Risk Management

(7) How are potential risks related to AI use (e.g. bias, transparency, regulatory compliance) assessed?

(8) Does your organization implement auditing or monitoring practices for algorithms? Who is responsible for this?

(9) How does data quality and governance impact AI-driven decision-making processes?

(10) What challenges do you face in ensuring the transparency and reliability of AI systems?

Section 4 – Organizational Culture and Training

(11) How is internal training on artificial intelligence managed? Does it involve only technical staff or other roles as well (e.g. HR, marketing)?

(12) Is there a structured training strategy, or are learning initiatives fragmented?

(13) Do you believe that the organization's personnel have the necessary tools to understand and manage AI-related implications?

(14) Are there internal experts who support colleagues in using or understanding AI systems?

Section 5 – Innovation and Future Vision

(15) How does your organization balance the need for technological innovation with the regulatory or ethical constraints associated with AI use?

(16) In your opinion, what are the main future opportunities and risks for organizations integrating AI into decision-making processes?

(17) What is your organization's strategic vision for AI adoption in the coming years?

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